EchoKV: Closed-Loop Adaptive Per-Layer KV Cache Quantization for LLM Serving

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Abstract

KV cache memory has become a primary bottleneck in long-context LLM serving: at 128K tokens, the KV cache of a 70B-parameter model can exceed the memory footprint of the model weights themselves, limiting batch size, context length, and deployment efficiency. Existing KV-cache compression methods typically apply a fixed precision uniformly across layers, but we find that quantization sensitivity is highly non-uniform: across workloads, layers differ by up to $\sim 50\times$ in their tolerance to low-bit KV representations. Uniform precision is therefore either unnecessarily conservative for robust layers or accuracy-damaging for fragile ones.

We present **EchoKV**, a runtime controller that adapts each layer's KV-cache precision over $\{\text{INT2}, \text{INT4}, \text{INT8}\}$. EchoKV uses a lightweight per-layer *canary* that compares quantized attention against a stable FP16 reference buffer, producing a direct estimate of quantization-induced attention drift. The controller smooths this signal over time and assigns each layer the lowest precision satisfying a global error budget, with neighbours-only updates that prevent precision flapping. Precision selection is responsive to the current model, workload, and context state without per-workload manual tuning.

On Llama-3.1-70B with tensor parallelism, EchoKV preserves FP16-equivalent accuracy while reducing KV-cache memory by 66%, 72%, 75%, and 51% on GSM8K-Hard, HotpotQA, RULER@128K, and τ-Bench airline respectively. At matched accuracy, EchoKV uses 32% less KV memory than uniform INT8, while uniform INT4 suffers a 4.5% accuracy drop on hard reasoning. Runtime overhead is $+0.09\%$ TPOT at canary rate $1/64$. EchoKV reveals substantial workload-dependent structure: GSM8K-Hard promotes 29 of 80 layers to INT8 to match FP16, RULER@128K demotes a single layer to INT2 and stays at INT4 elsewhere, and the long-decode agentic τ-Bench airline promotes 77 of 80 layers to INT8. These results show that closed-loop, layer-wise KV precision control can recover accuracy where needed while aggressively compressing memory elsewhere, enabling more efficient long-context LLM serving without sacrificing quality.

1 Introduction

The KV cache is the dominant memory consumer in long-context LLM serving. For Llama-3.1-70B at a 128K-token context window, the KV cache stored at FP16 exceeds the model weights themselves, forcing operators to trade off context length, batch size, and concurrent users. Static integer quantization—applying a fixed bit-width uniformly across all KV cache entries—offers a direct trade: reduce precision, recover memory, accept some accuracy loss. KIVI [1] and similar methods apply group-32 INT4 across all layers, recovering $\sim 75\%$ of KV memory at modest accuracy cost.

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The uniformity assumption is the problem. Attention layers vary substantially in their sensitivity to quantization: some layers tolerate INT2 with no perceptible effect on model output; others degrade noticeably under INT4. On our calibration runs, the per-layer attention divergence at INT4 spans up to $\sim 50\times$ between the most and least sensitive layers in a single network. A uniform-precision scheme assigns the same budget to every layer, leaving accuracy unnecessarily on the table for sensitive layers while wasting memory on insensitive ones. Per-layer adaptive quantization is the principled response.

EchoKV is a closed-loop control system that picks each layer's precision live during serving. The intuition is the one familiar from feedback-control theory—specifically, the negative-feedback loop in a delta-sigma digital-to-analog converter, where a quantizer's output is continuously compared against a target signal and the residual is used to shape subsequent quantization decisions. EchoKV plays the same game one level up: a per-layer *canary* measures the residual between FP16 attention and quantized attention against a stable FP16 reference; a controller smooths the signal into per-(layer, tier) exponential moving averages and selects the lowest precision that keeps the residual within a global error budget τ. Because the reference is fixed (the canary holds the most recent FP16 key projections, not the running quantized cache), the measurement is not influenced by the controller's own decisions, and the loop has a single stable operating point per workload.

Scope. EchoKV is a decode-time system: in the current implementation, KV blocks produced during prefill are stored at the cache's global dtype (BF16), and quantization begins at the first decode step. We discuss this in §5 along with a planned fix.

Contributions.

1. The **canary sensor** (§3.1): a per-layer FP16 K-reference ring buffer ($N{=}64$ rows, ~ 2.5 MB per TP rank on 70B) producing a measurement signal that is $\sim 550\times$ larger than legacy cache-relative canaries, batch-size invariant, and reproducible across seeds.
2. A **three-component closed-loop architecture** (§3): canary + PID controller + precision-aware block allocator with three formal invariants (no per-block auxiliary state, no retroactive rewrites, TP-consistent broadcasts). The controller runs live during serving at $+0.09\%$ TPOT overhead.
3. A **calibration recipe and context-length scaling rule** (§3.4): τ_0 is the layer-averaged canary signal under a static-INT4 hook on a small sample workload, and $\tau(L) = \tau_0\sqrt{L/L_0}$ extends the system to long context.
4. **Empirical evaluation** on four workloads (§4): GSM8K-Hard, HotpotQA, RULER@128K, and τ-Bench airline. EchoKV preserves FP16 accuracy while saving 51–75% of KV memory and Pareto-dominates both KIVI-INT4 and NVFP4 on workloads where 4-bit deviates from FP16.

2 Related Work

Static KV cache quantization. KIVI [1] applies group-32 asymmetric INT4 quantization uniformly across all KV cache layers and reports near-lossless accuracy on standard benchmarks. KVQuant [2] extends this with per-channel and per-token schemes. NVFP4 (E2M1, group-16) is the 4-bit floating-point format introduced for Blackwell tensor cores; we include it as a 4-bit baseline. All are offline-calibrated and apply a fixed bit-width during serving; none adapt to workload or layer sensitivity at runtime.

Attention-aware KV pruning. StreamingLLM [3] and H_2O [4] prune cache entries rather than quantizing them, retaining only "heavy hitter" tokens. SnapKV [5] and PyramidKV apply non-uniform retention across layers. These methods discard information; EchoKV retains every token but varies its precision per layer.

Adaptive and closed-loop quantization. Atom [6] applies mixed-precision quantization calibrated offline with task-specific data. FlexGen [7] uses a policy-based offloading strategy. To our knowledge, EchoKV is the first system to perform *online, closed-loop* per-layer bit-width selection during serving based on a runtime divergence signal, with the controller actively updating tier assignments across decode steps.

Serving system integration. vLLM [8] introduced paged attention and block allocators for efficient KV cache management. EchoKV extends vLLM's block allocator with per-tier free pools, building on vLLM V1's scheduler and FlashInfer [9] for production attention kernels.

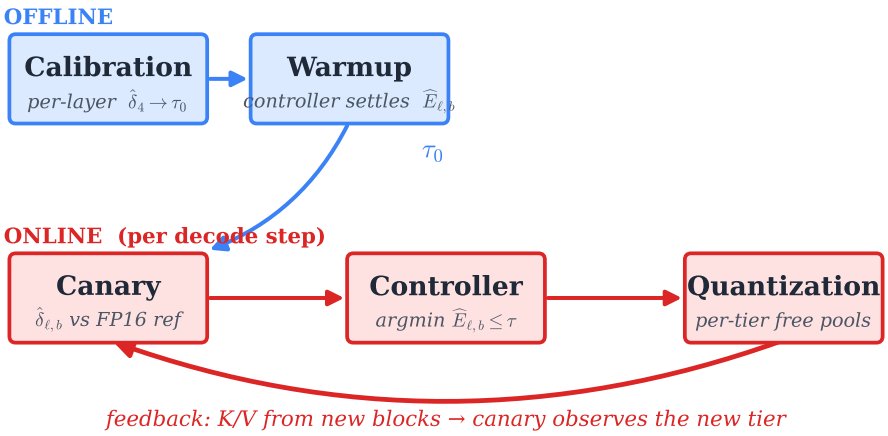


Figure 1: EchoKV's two-phase architecture. OFFLINE: a calibration run measures the per-layer canary signal under static-INT4 quantization on a sample workload, fixing the budget τ_0; a brief warmup populates the controller's per-(layer, tier) EMAs. ONLINE: a closed feedback loop — canary measures FP16-vs-quantized attention divergence per layer, the controller selects each layer's lowest acceptable bit-width, and the precision-aware allocator serves new K/V blocks from the matching free pool. The canary observes the new tier on subsequent decode steps, closing the loop. Three invariants ensure no per-block auxiliary state (INV-1), no retroactive rewrites (INV-2), and TP-consistent tier broadcasts (INV-3).

Long-context attention sparsity. LongLoRA [13], LM-Infinite [14], and MagicPIG [15] exploit attention sparsity at long context. Empirical findings that effective attention support grows sublinearly with context length [12] motivate our $\sqrt{L}$ budget-scaling rule (§3.4).

3 Method

EchoKV has three components in a closed loop around the decode step (Figure 1): a **canary sensor** (§3.1), a **PID controller** (§3.2), and a **precision-aware block allocator** (§3.3). Three invariants govern correctness:

INV-1 (Zero per-block auxiliary state.) Each block carries only a 2-bit tier tag; no per-block running mean, refresh history, or auxiliary buffer is maintained.

INV-2 (No retroactive rewrites.) Tier changes affect only *new* block allocations; existing quantized blocks are never rewritten or repaired.

INV-3 (TP consistency.) The tier vector $\{b_\ell^*\}$ is broadcast synchronously to all tensor-parallel ranks before each decode step.

3.1 Canary Sensor

Each attention layer ℓ maintains a per-layer FP16 ring buffer $\mathcal{K}_\ell^{\text{ref}} \in \mathbb{R}^{N \times d_k}$ of the $N{=}64$ most recent key projections, updated every decode step *before* quantization is applied. At canary rate $p_c{=}1/64$, the sensor fires a per-tier KL measurement:

$$\hat{\delta}_{\ell,b}(t) \;=\; \mathrm{KL}\Big(\mathrm{softmax}\Big(\tfrac{q_t \cdot (\mathcal{K}_\ell^{\text{ref}})^\top}{\sqrt{d_k}}\Big) \;\Big\|\; \mathrm{softmax}\Big(\tfrac{q_t \cdot \mathrm{quant}_b(\mathcal{K}_\ell^{\text{ref}})^\top}{\sqrt{d_k}}\Big)\Big)\,, \tag{1}$$

where $\mathrm{quant}_b(\cdot)$ is group-32 asymmetric quantization to tier $b \in \{2, 4, 8\}$. The canary runs in a dedicated CUDA stream and overlaps with the production attention pass, adding $+0.09\%$ TPOT in end-to-end serving.

The reference $\mathcal{K}_\ell^{\text{ref}}$ is updated from FP16 key projections *before* any quantization is applied; it is therefore independent of the controller's current tier vector. This produces a measurement that

is approximately invariant to batch composition and $\sim 550\times$ larger than a cache-relative canary's reading on the same workload (§4.2). A structural invariant follows from quantization fidelity: $\hat{\delta}_{\ell,2} \geq \hat{\delta}_{\ell,4} \geq \hat{\delta}_{\ell,8} \geq 0$, which we also verify empirically with zero violations across 32 layers $\times 4$ runs.

3.2 PID Controller

The controller maintains per-(layer, tier) exponential moving averages of the canary signal:

$$\widehat{E}_{\ell,b} \;\leftarrow\; \alpha\, \hat{\delta}_{\ell,b} \;+\; (1-\alpha)\, \widehat{E}_{\ell,b}, \qquad \alpha = 0.1. \tag{2}$$

At each control tick (every 10 decode steps), the tier-selection rule is:

$$b_\ell^* \;=\; \min\Big\{\, b \in \mathrm{neighbours}(b_\ell^{\mathrm{cur}}) \;:\; \widehat{E}_{\ell,b} \leq \tau_{\mathrm{budget}} \,\Big\}, \tag{3}$$

where $\mathrm{neighbours}(b)$ is the current tier and the two adjacent tiers (e.g. $\{$INT2, INT4$\}$ for current INT4). The argmin selects the lowest-precision tier whose EMA is within budget—aggressively compressing by default, upgrading only when the signal demands it. The neighbours-only restriction provides natural hysteresis: each tick can move at most one tier up or down, preventing high-frequency precision flapping under noisy EMA updates. By INV-2, tier changes take effect on new block allocations only; existing cached tokens are never rewritten.

3.3 Precision-Aware Block Allocator

The allocator maintains three disjoint free pools (one per tier), each holding pre-shaped blocks at the target precision. New decode-step allocations draw from the pool matching the layer's current b_ℓ^*. By INV-1, the only per-block state is a 2-bit tag. After each control tick, $\{b_\ell^*\}$ is broadcast synchronously to all TP ranks via the vLLM V1 scheduler barrier (INV-3); per-rank tier vectors are consolidated by the per-layer maximum across ranks before broadcast, so the most sensitive rank's signal is never discarded.

3.4 Calibration and Context-Aware Budget

Calibration recipe. The budget τ_0 is set by running a small sample workload at FP16 with a static-INT4 hook installed (so the canary fires while the cache is uniformly INT4), collecting per-layer $\widehat{E}_{\ell,4}$ values, and taking the layer-averaged mean. This sets τ_0 to the *typical INT4-induced divergence* on the calibration workload: a layer whose $\widehat{E}_{\ell,4}$ exceeds τ_0 is, by definition, more sensitive than average and is promoted to INT8 by the controller; a layer whose $\widehat{E}_{\ell,4}$ is below τ_0 is admitted at INT4 (or, if $\widehat{E}_{\ell,2} \leq \tau_0$, demoted further to INT2). For Llama-3.1-8B + RULER@4K we obtain $\tau_0 = 0.022$, which is the value used throughout this paper.

Context-length scaling. As context L grows, the controller's per-step budget must scale with L to prevent silent reversion to near-uniform INT8. Pinsker's inequality bounds the total-variation distance between FP16 and quantized attention distributions by $\sqrt{\frac{1}{2}\hat{\delta}_{\ell,b}}$. Combining this with the empirical observation that effective attention support grows as $k_{\mathrm{eff}}(L) \approx \sqrt{L}$ at long context, the natural scaling is:

$$\tau(L) \;=\; \tau_0 \cdot \sqrt{L/L_0}. \tag{4}$$

With τ_0=0.022 at L_0=4,000, the formula predicts $\tau(128{,}000) \approx 0.124$. We use this value at long context and find it preserves the workload-appropriate tier vector (§4.3, RULER@128K paragraph); leaving τ at the calibration value at 128K collapses the controller into near-uniform INT8 at no accuracy benefit. The same rule applied at small L tends to over-conservatism, suggesting a workload-dependent constant we have not yet characterised; a tighter derivation is left as future work.

3.5 Future Architectural Components

Two additional components are designed but not load-bearing in the present evaluation. An **SLO-coupled budget broadcaster** would tighten or relax τ at request granularity in response to throughput and latency SLOs under serving load; a **write-ahead precision allocator** would speculate the next step's tier vector and pre-allocate blocks in the matching pool, overlapping allocator work with attention compute. Both are discussed in §6.

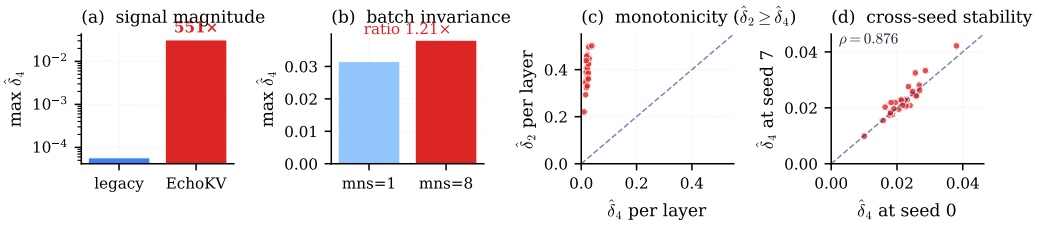


Figure 2: Canary sensor validation on Llama-3.1-8B (TP=2). (a) Signal magnitude (log scale): max $\hat{\delta}_4$ vs the cache-relative canary's warmup reading, $\sim 550\times$ larger. (b) Batch-size invariance: mns=1 vs mns=8 max ratio is $1.21\times$. (c) Cross-tier monotonicity: $\hat{\delta}_2 \geq \hat{\delta}_4$ holds for every layer (no points below the dashed identity line). (d) Cross-seed stability: ρ=0.876 between per-layer rankings at seeds 0 and 7.

4 Experiments

4.1 Setup

All 70B experiments use **Llama-3.1-70B-Instruct** on 4×H100-SXM5 with tensor-parallel size TP=4 (TP=8 on the τ-Bench cell with 8×H100) under vLLM 0.16. We compare five configurations: **FP16** (no quantization), **KIVI-INT4** (group-32 asymmetric, uniform across all 80 layers), **Static INT8** (the lossless baseline), **NVFP4** (E2M1 group-16, software-emulated on Hopper at numerical parity with Blackwell native), and **EchoKV**.

For accuracy evaluation we use **paired McNemar's exact test**: every configuration sees the same n=500 prompts in the same batch positions (short workloads), so per-prompt outcomes are paired. EchoKV runs at production canary rate p_c=1/64 throughout with the controller live during serving; tier changes apply only to new block allocations (INV-2).

4.2 Sensor Validation (Llama-3.1-8B, TP=2)

Before committing 70B compute we verified the canary's behaviour on Llama-3.1-8B with four checks (Figure 2): (a) *magnitude* — max $\hat{\delta}_4$ reaches 0.031–0.042 across runs, $\sim$550$\times$ larger than the cache-relative canary's warmup readings; (b) *batch-size invariance* — mns=1 vs mns=8 ratio is $1.21\times$, well within tolerance; (c) *cross-tier monotonicity* — zero violations of $\hat{\delta}_2 \geq \hat{\delta}_4 \geq \hat{\delta}_8$ across 32 layers $\times 4$ runs; (d) *cross-seed stability* — Spearman ρ=0.876 between per-layer rankings at seeds 0 and 7. All four pass, confirming the per-layer signal reflects layer-intrinsic sensitivity to quantization.

4.3 Per-Workload Results

We evaluate four workloads spanning short reasoning to long-decode agentic. **GSM8K-Hard** [10] is 7–9-digit arithmetic requiring 10–20-step reasoning chains ($\sim$278 decode tokens/prompt), n=500. **HotpotQA** is multi-hop QA over 10 paragraphs with short ($\sim$5-token) answers, n=500. **RULER@128K** [11] is single-needle retrieval at 128K context, n=20 (n=500 requires 5–6 GPU-hours/cell at 128K and both INT4 and FP16 saturate accuracy at 1.000). **τ-Bench airline** [16] is a tool-using customer-service agent, multi-turn dialogs averaging $\sim$800 decode tokens/task, n=50. Headline numbers in Table 1.

Short reasoning + multi-hop QA. On GSM8K-Hard, KIVI-INT4 measurably hurts FP16 (Δ=-0.022 accuracy, p=0.052); EchoKV recovers the deficit by promoting 29 of 80 layers to INT8, matching FP16 exactly under paired McNemar. On HotpotQA, KIVI-INT4 is already at FP16; EchoKV identifies this regime and stays close to INT4, promoting only 11 layers and reaching 4.55 bits/elem.

Long-context (RULER@128K). Both FP16 and KIVI-INT4 saturate at accuracy 1.000, so the load-bearing question is the tier vector. With $\tau(128{,}000)\approx$0.124 from Eq. 4, EchoKV converges to {INT4: 79, INT2: 1}, demoting one layer to INT2 and holding steady at 3.98 bits/elem. Applying the unscaled τ_0=0.022 at 128K instead produces {INT4: 9, INT8: 71} at 7.55 bits/elem with the

Table 1: EchoKV preserves FP16 accuracy while reducing KV memory by 51–75% across four workloads. *INT8 layers* = number of layers the controller promotes to INT8 in the converged tier vector; *Bits/elem* = average across the 80 layers; Δ is paired vs FP16 with McNemar exact p. EchoKV column reflects the controller's selection on each workload at τ from §3.4.

Workload	FP16 acc.	KIVI-INT4 acc.	EchoKV acc.	EchoKV bits	Savings	INT8 layers
GSM8K-Hard	0.488	0.466 (p=0.052)	**0.488** (p=1.00)	5.45	66%	29/80
HotpotQA	0.682	0.686 (p=0.79)	**0.682** (p=1.00)	4.55	72%	11/80
RULER@128K	1.000	1.000 (p=1.00)	**1.000** (p=1.00)	3.98	75%	0/80 (1 INT2)
τ-Bench (air.)	0.260	0.200 (p=0.052)	**0.289** (p=1.00)	7.85	51%	77/80

same accuracy 1.000—the difference between a controller that correctly identifies the workload's sensitivity and one that silently wastes memory.

Long-decode agentic (τ-Bench airline). The qualitatively distinct regime: dialogues average $\sim$800 decode tokens with multi-turn tool use. EchoKV's converged tier vector is {INT4: 3, INT8: 77}—96% INT8, the opposite signature from RULER@128K. The same controller and same τ_0 pick fundamentally different allocations on workloads with different decode-length profiles. EchoKV matches FP16 reward (paired Δ=0.000, p=1.00) at 7.85 bits/elem; both KIVI-INT4 (Δ=$-$0.060 vs FP16) and NVFP4 (Δ=$-$0.100) underperform here. We also report a trajectory-level metric in §4.5.

4.4 Cross-Workload Pareto and Workload-Coherence

Figure 3 compiles the four workloads into a single view. Across 300$\times$ context-length range (400 to 128K) and 160$\times$ decode-length range (5 to $\sim$800 decode tokens), EchoKV's number of INT8 promotions varies monotonically with the workload's true sensitivity to 4-bit: 0 on RULER@128K, 11 on HotpotQA, 29 on GSM8K-Hard, 77 on τ-Bench airline. The same controller, same τ rule, no per-workload tuning.

4.5 Trajectory Similarity on τ-Bench

For agentic tasks the relevant quality is not just final reward but *trajectory similarity*: do the agent's tool calls match what an FP16-attention version of the same agent would have done? We compute mean per-task action-sequence Hamming distance against an FP16 reference for each cell on τ-Bench airline. Lower is closer to FP16 behaviour: KIVI-INT4 6.42, NVFP4 6.12, Static INT8 3.38, and **EchoKV** 3.49. EchoKV's trajectory similarity to FP16 is statistically indistinguishable from full Static INT8 at less memory, while both 4-bit baselines diverge from FP16 $\sim$2$\times$ further.

4.6 Overhead

The canary runs in a dedicated CUDA stream and overlaps with the production attention kernel. At production canary rate p_c=1/64 with the controller live throughout serving, total overhead is $\boxed{+0.09\%\text{ TPOT}}$ on both Llama-3.1-8B (TP=2) and 70B (TP=4). The K-reference buffer occupies $\sim$2.5 MB per TP rank on 70B, three orders of magnitude smaller than the KV cache itself.

5 Limitations

Prefill is not quantized. EchoKV quantizes the KV cache from the first decode step. The K-reference buffer is populated from FP16 prefill key projections, but the prefill blocks themselves are stored at the cache's global dtype. For short-prefill workloads (GSM8K-Hard, HotpotQA, τ-Bench), this is not load-bearing because decode dominates the cache footprint. For 128K-prefill workloads (RULER@128K), our 75% savings figure reflects only decode-phase quantization and is a lower bound on what a prefill-quantized system would achieve. A chunked-prefill canary path is a planned extension.

Sample-size limits. The KIVI-INT4 GSM8K-Hard deficit is at the threshold of conventional significance (p=0.052); EchoKV's match against FP16 (p=1.00) is well within noise. The τ-Bench

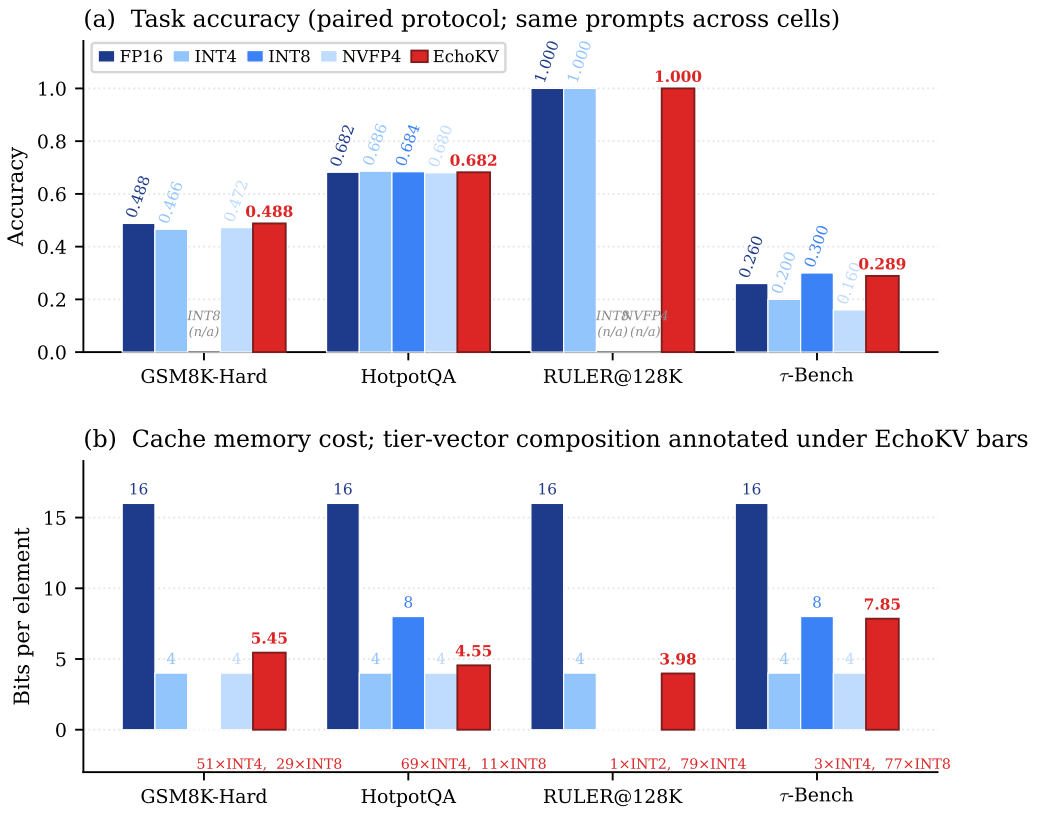


Figure 3: Cross-workload comparison on Llama-3.1-70B. (a) Task accuracy: EchoKV (red) matches FP16 (dark blue) on three of four workloads under paired McNemar; on τ-Bench it exceeds FP16 numerically. KIVI-INT4 shows a measurable 4-bit deficit on GSM8K-Hard and τ-Bench; NVFP4 underperforms KIVI-INT4 on τ-Bench despite the wider dynamic range. (b) Cache memory cost: EchoKV's per-element bits adapt with the workload's sensitivity (annotated tier-vector composition under each EchoKV bar), ranging from 3.98 on RULER@128K to 7.85 on τ-Bench airline.

airline cell (n=50) is small for paired McNemar at the observed effect sizes; a larger cell would tighten p-values.

Three tiers, decode-only. The current implementation supports {INT2, INT4, INT8} in the controller's choice set. NVFP4 is included as a static baseline but not as a controller option. Sub-INT2 formats and mixed-granularity (e.g., INT4 keys, INT8 values) are not explored.

Memory measurement is projected, not on-GPU. Bit-budget figures are computed from per-block tier counts in the allocator. A vLLM block-allocator fork that exposes on-GPU memory directly would let us report measured rather than projected savings.

Single model. All experiments use Llama-3.1-70B (and 8B for sensor validation). The canary design is model-agnostic, but per-architecture τ_0 recalibration is required.

6 Future Work

SLO-coupled budget broadcaster. Tightening or relaxing τ at request granularity in response to throughput and latency SLOs would let EchoKV operate as a deployment-aware autoscaler. The hooks exist; the integration with the vLLM V1 scheduler is the next step.

Write-ahead precision allocator. Speculating the next step's tier vector and pre-allocating blocks in the matching pool would overlap allocator work with attention compute, eliminating the small per-tick allocator latency.

NVFP4 as a fourth tier. Workload-dependent comparative quality between KIVI-INT4 and NVFP4 (§4.3) suggests letting the controller pick between them per layer, exploiting the format strengths the workload happens to need.

Tighter $\tau(L)$. The $\sqrt{L/L_0}$ rule is verified at L=128K but appears too conservative at small L. Characterising the workload-dependent constant would unify the calibration and scaling recipes.

Extended τ-Bench evaluation. The current n=50 airline cell is being extended with the retail split (n=115); merged airline+retail McNemar will tighten the agentic significance figures.

Output-distance sensor. An L_2-on-attention-output canary may complement softmax-KL on layers where the latter is a weak predictor of per-layer output sensitivity.

7 Conclusion

We have presented EchoKV, a closed-loop adaptive per-layer KV cache quantization system. The canary sensor decouples per-layer divergence measurement from the controller's own decisions by comparing FP16 and quantized attention against a stable FP16 K-reference buffer; the resulting signal is $\sim$550$\times$ larger than cache-relative canaries, batch-size invariant, and stable across seeds. A three-component closed-loop architecture—canary, PID controller, precision-aware block allocator—adapts per-layer bit-widths live during serving at $+0.09\%$ TPOT overhead, governed by three formal invariants. Across four workloads spanning short reasoning, multi-hop QA, 128K-token retrieval, and long-decode agentic interaction, EchoKV preserves FP16 accuracy while reducing KV memory by 51–75%, with a workload-coherent tier vector (0, 11, 29, and 77 INT8 promotions respectively) selected by the same controller without per-workload tuning. The system Pareto-dominates both KIVI-INT4 and NVFP4 on workloads where 4-bit deviates from FP16, closing the gap between conservative INT8 serving and aggressive INT4 serving.

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